Liner Xiang

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EDUCATION

University of California, Irvine Sep 2022-June 2027(expected)
School: Department of Statistics, Donald Bren School of Information and Computer Sciences
Degree Expected: Ph.D. in Statistics
Co-advisors: Hengrui Cai and Weining Shen
GPA: 3.99/4.0

University of Science and Technology of China Sep 2018-June 2022
School: School of the Gifted Young
Degree: B.S. in Statistics

AWARDS

NeurIPS Travel Award Oct 2026
Statistics Travel Award, UC Irvine Oct 2026
Outstanding Graduate Teaching Assistant Award, Department of Statistics, UC Irvine Sep 2025
MBZUAI AI Forum Travel Award May 2025
Excellent Student Scholarship, USTC Sep 2018, 2019, 2020, 2021

RESEARCH INTERESTS

My research lies at the intersection of reinforcement learning, natural language processing, and causal inference. I develop principled statistical and optimization frameworks for complex environments, with a focus on sequential decision-making under interference and strategic interactions. By integrating reinforcement learning, game theory, and statistical learning, I develop scalable algorithms with rigorous theoretical guarantees. On the applied side, I focus on evaluating and improving large language models (LLMs) under weak supervision and distribution shift, as well as developing machine learning methods for precision medicine. My work emphasizes off-policy evaluation, reinforcement learning from human feedback, robust routing, cost-efficient LLM-as-a-judge systems, and adaptive decision-making for personalized treatment strategies.

PUBLICATIONS / ACCEPTED

* Equal contribution

Zhang, L.*, **Xiang, L.***, Zhang, W.*, Zheng, Y., and Cai, H. (2026).
Large Language Model Alignment with Complex Feedback: A Survey. **(Survey)**
Findings of the **European Chapter of the Association for Computational Linguistics (EACL) 2027**.

Xiang, L., Zhang, W., and Cai, H. (2026).
OTROPE: Optimal Transport-based Robust Off-policy Evaluation for Large Language Models. **(OTROPE)**
Advances in Neural Information Processing Systems (NeurIPS), 2026.

Zhang, W.*, Zhang, L.*, **Xiang, L.***, and Cai, H. (2026).
Reasoning Is Not Free: Robust Adaptive Cost-Efficient Router for LLM-as-a-Judge. **(RACER)**
International Conference on Machine Learning (ICML), 2026.

Ding, X.*, Meng, Y.*, **Xiang, L.***, and Boden-Albala, B. (2024).
Stroke Recurrence Prediction Using Machine Learning and Segmented Neural Network Risk Factor Aggregation. **(SNA)**
Discover Public Health, 21(1).

UNDER REVISION

Xiang, L., Wang, Y., and Cai, H. (2026).
Policy Optimization and Statistical Inference for Online Contextual Matrix Games. **(OnGameLearn)**
Major revision (submitted) at **Journal of the American Statistical Association (JASA)**.

Xiang, L., Wang, J., and Cai, H. (2026).
Foresighted Online Policy Optimization with Interference. **(FRONT)**
Accepted at the **Causal Reinforcement Learning Workshop (CRLW), RLC 2025.**
Invited revision at **Journal of Machine Learning Research (JMLR).**

Alantari, H., **Xiang, L.**, Cai, H., Shen, W., and Currim, I. (2026).
Modeling Book Lovers' Sequential Decisions to Read, Rate, and Review: Missing Data Imputation for Efficient Estimation.
Major revision (submitted) at **European Journal of Marketing**.

UNDER REVIEW

Xiang, L., Jiang, H., Liu, Y., Jiang, J., Cai, H., and Song, R. (2026).
Counterfactual Outcome Prediction in Foundation Models: An Inverse Probability Weighting Approach.
Under review at **Association for Computational Linguistics Rolling Review (ARR), 2026**.

Zhuang, X.**, Xiang, L.,** and Zhou, W. (2026).
When and How to Collaborate: State-Adaptive Sequential Multi-Agent Collaboration via Reinforcement Learning.
Under review at **Association for Computational Linguistics Rolling Review (ARR), 2026**.

Gui, Z., **Xiang, L.**, Guo, J., and Shen, W. (2026)
Is It the Edit or the Rerun? Decomposing Recorded Repairs in Reflective Prompt Optimization.
Under review at **Association for Computational Linguistics Rolling Review (ARR), 2026**.

Dai, J., Ding X., **Xiang, L.**, Meng, Y., Kim, Y., Boden-Albala, B., and Jiang, L. (2026)
Machine Learning-based Prediction of Incident Stroke in Adults with Diabetes Using Large-scale Electronic Health Record Data
Under review at **JMIR Formative Research.**

INDUSTRY EXPERIENCES

Applied Scientist | Amazon Seattle, WA
Mentors : Yang Liu, Jieyi Jiang, and Rui Song Jun 2026-Sep 2026

- Developed a causal fine-tuning framework for pretrained sequence models to improve counterfactual outcome prediction under confounded sequential treatment assignment.
- Designed an inverse-propensity-weighted training objective with joint propensity and outcome modeling to correct behavior-policy selection bias and improve learning for underrepresented treatment–context combinations.
- Evaluated the model-agnostic framework on real-world data across multiple sequence-modeling backbones, demonstrating improved counterfactual prediction performance over standard fine-tuning.

Statistician | Quantum Leap Healthcare Collaborative San Francisco, CA
Mentors: Peter Norwood and Philip Beineke Jun 2025-Sep 2025

- Contributed to the design of the I-SPY2 adaptive breast cancer trial using Bayesian modeling and reinforcement learning.
- Built an interactive trial-planning tool to help clinicians compare adaptive treatment strategies before deployment.

TEACHING

Guest lecturer for graduate course at UC Irvine:
BATS 209A: Introduction to Medical Statistics (Fall 2026)

Teaching fellow for graduate course at UC Irvine:
DATA 200AP: Intermediate Probability and Statistical Theory (Fall 2024)

Teaching fellow for undergraduate course at UC Irvine:
STATS 7: Basic Statistics (Winter 2024, Fall 2025)
STATS 8: Introduction to Biological Statistics (Winter 2023, Fall 2023)
STATS 67: Introduction to Probability and Statistics for Computer Science (Fall 2023)
STATS 111/202: Statistical Methods for Data Analysis II (Winter 2025)
STATS 115: Bayes Data Analysis (Winter 2026)
STATS 120C/281C: Introduction to Probability and Statistics III (Spring 2023, Spring 2024, Spring 2025, Spring 2026)

Teaching fellow for undergraduate course at USTC:
MATH1009: Linear Algebra B1 (Fall 2021)

ACADEMIC SERVICE

Reviewer:
ICLR (2025–2027), ICML (2026), NeurIPS (2026), ACL Rolling Review (2026), Biostatistics, The Annals of Applied Statistics, Biometrics.

Membership:
American Statistics Association (2024-Present), Eastern North American Region International Biometric Society (2025-Present).

Invited Events:
MBZUAI AI Forum (May 2025), Global Intern Day – Amazon Social (Jul 2026).

TALKS&PRESENTATIONS

Poster Presentation, NeurIPS 2026 Dec 2026
OTROPE: Optimal Transport-based Robust Off-policy Evaluation for Large Language Models

Internship Presentation, Amazon Sep 2026
Causal Fine-Tuning for Foundation Models: Leveraging Causal Structure in Sequential Data for Outcome Simulator

Research Presentation, UC Irvine Oct 2025
Reliable Evaluation of LLMs through Rectified Estimation

Internship Presentation, QLHC Sep 2025
Sequential Decision-Making for Adaptive Clinical Trial Design

Advancement to Candidacy Presentation, UC Irvine Jun 2025
Decisions in the Wild: Foresight, Adversaries, and Human Feedback in Online Policy Optimization and Evaluation